Lab 4 — Tracing the Compute Plane

Description

You will start a classic compute resource in the training workspace and watch the Azure resources it creates appear and disappear. Keep the Databricks workspace open in one browser tab and the Azure portal in another so you can switch between them.

Learning Objectives

- Create and start an all-purpose compute resource within the limits set by a cluster policy.
- Identify the Azure resources a running cluster creates, and where they appear.
- Use the compute detail page to inspect configuration, event log and driver logs.
- Explain how classic compute differs from serverless compute in what it deploys.

Step by Step Guide

1. Open Compute. Click Compute in the sidebar and select the All-purpose compute tab. Note any existing compute resources and their state.
2. Start a new compute resource. Click Create compute. Note the Policy selector at the top of the form — the policy limits which options you can change.
3. Configure it. Give it a name including your own username. Choose the latest LTS Databricks Runtime. In the Performance section, set a small worker count and an auto termination period of 20 minutes. Leave Access mode as Auto.
4. Note the estimate. Before creating, review the DBU estimate shown on the form. This is the Databricks charge only — the underlying Azure VMs are billed separately.
5. Create the compute resource. Click Create and note the time. Starting takes several minutes; move to the Azure portal while it runs.
6. Watch the Azure resources appear. In the Azure portal, open the managed resource group and refresh it every minute. Virtual machines, managed disks and network interfaces appear as the cluster starts. Count them against your driver and worker configuration.
7. Find where the network interfaces sit. Open one of the network interfaces and note which virtual network and subnet it is attached to. This is the compute plane running inside the network you reviewed in Session 3.
8. Review the compute detail page. Back in Databricks, open your compute resource and work through the tabs — Configuration, Libraries, Event log, Spark UI, Driver logs and Metrics. Open the Event log and read the start sequence.
9. Run something on it. Create a notebook in your home folder, attach it to your compute resource and run:

```
SELECT current_user() AS user, current_catalog() AS catalog;
```

```
SHOW SCHEMAS IN actuarial;
```

10. **Terminate the compute resource.** Return to the compute page and click Terminate. Do not skip this step — the Azure VMs are billed for as long as they run.
11. **Watch the Azure resources disappear.** Refresh the managed resource group. The virtual machines, disks and network interfaces are removed. The workspace storage account remains, because it belongs to the workspace rather than to the cluster.
12. **Contrast with serverless.** Serverless compute runs in a Databricks-managed subscription, so no equivalent resources appear in your managed resource group at all. Note what that changes for network control and cost visibility.

Outcome

You have seen the compute plane created and destroyed on demand, and can point to the exact Azure resources a running cluster consumes. You can locate a cluster's configuration, event log and driver logs, and explain why classic and serverless compute present very different pictures in the Azure portal.